A small supervision budget barely improves weak-to-strong generalization

On GPT-2 / BoolQ, mixing a ground-truth budget into the weak labels helps only modestly — and *where* and *how* it is spent show little effect.

The student largely reproduces the teacher's errors, and ground truth corrects mainly the examples it directly labels — so the effective axis is *volume*, not placement or combination.

Mayank Mahajan

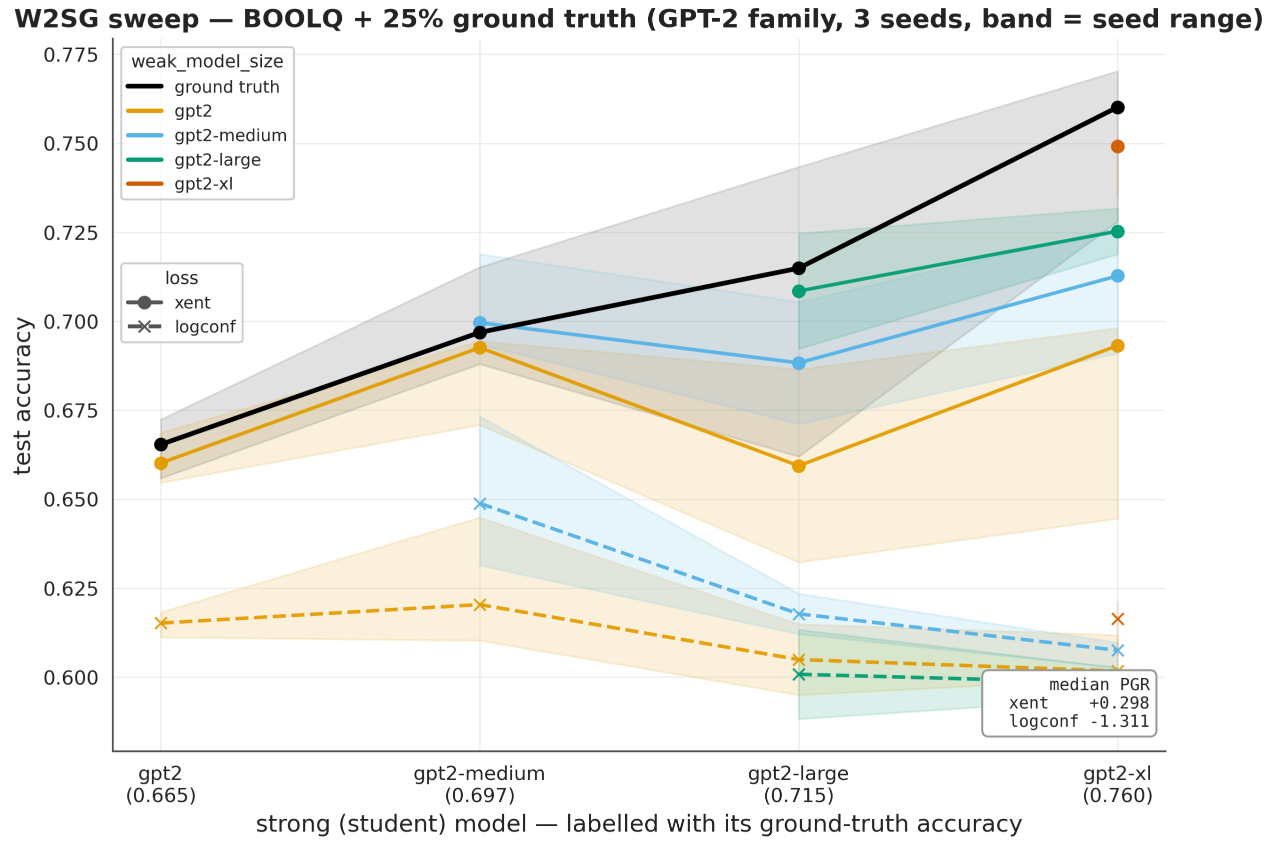
Scope & setup

- **Setting:** weak-to-strong generalization — a strong *student* is trained on a weak *teacher's* labels.
- **Extension:** a *supervision budget* — mix a fraction of ground-truth ("strong") labels into the weak labels, and sweep that fraction.
- **Models: GPT-2 family only** — gpt2 / medium / large / xl, with within-family student–teacher pairs.
- **Tasks: BoolQ** (required) + **SciQ** (cross-task check).
- **Readout:** median **PGR** across the model sweep (raw accuracy primary; PGR secondary).
- **Deliberately held fixed:** larger capability gaps, generative / reward-modeling tasks, other architectures, training schedule (see next steps).

3 seeds · paired per-(pair,seed) contrasts · advance predictions · gpt2-large (seed 1) excluded by rule

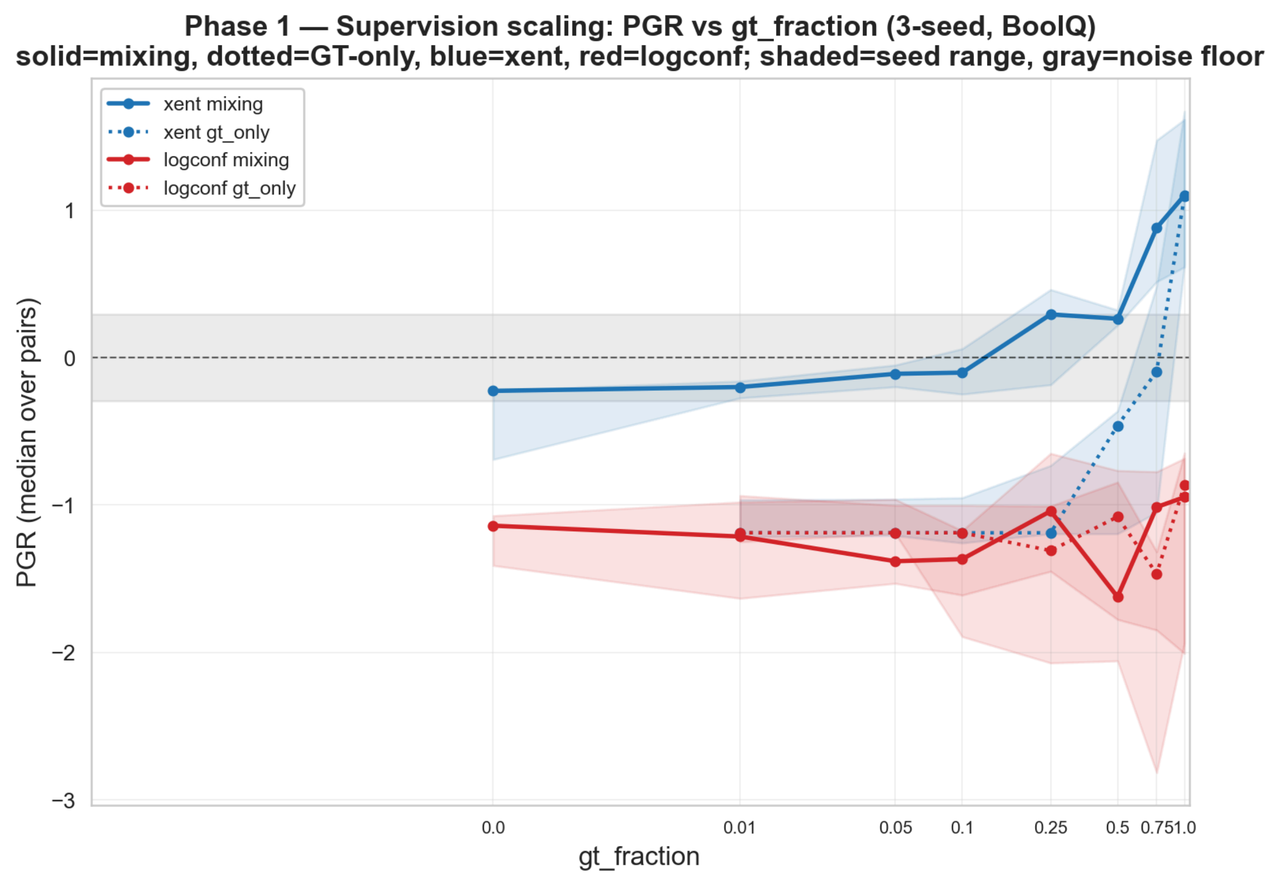
GPT-2 sweep at 25% ground truth

- GPT-2 family, BoolQ, **25% ground truth mixed into the weak labels** — standard sweep format.
- **Median PGR = +0.30** with cross-entropy training; the confidence loss (dashed) sits well below it.
- The 0% baseline is **−0.22**, making the +0.30 a **small** positive shift.



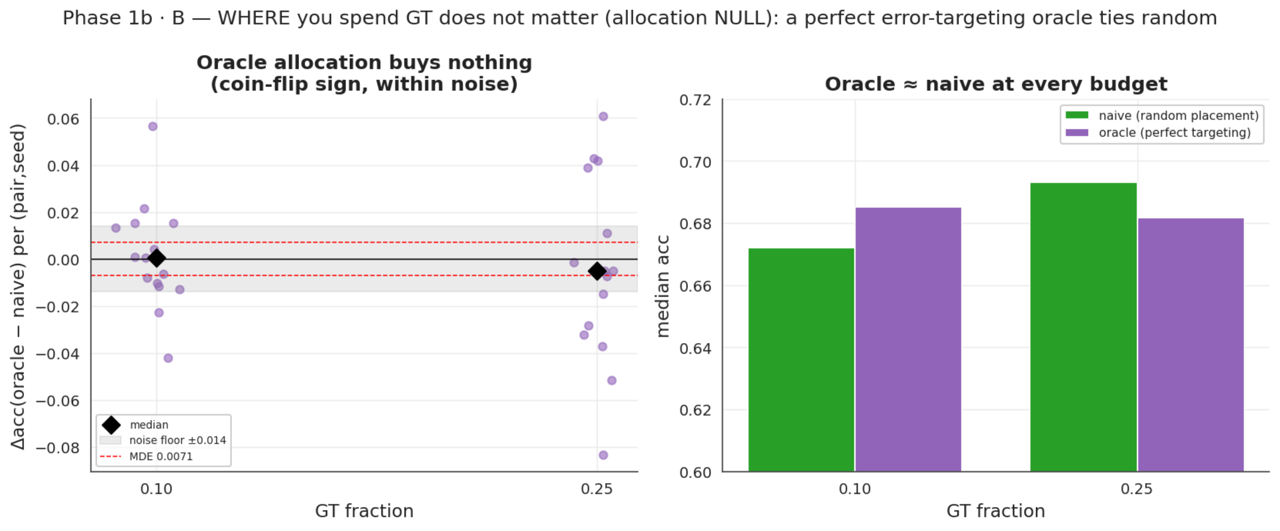
How *much*? — back-loaded, and saturating by ~75%

- Up to 10% GT, the gain over the 0% baseline stays within the 0.014 noise floor.
- Median PGR is back-loaded: −0.22 → +0.30 (0.25) → +0.28 (0.50) → +0.90 (0.75) → +1.04 (1.0).
- The 0.75 point: the 0.50→0.75 step is the largest, and 0.75→1.0 is within noise.
- A prior prediction of a concave knee at 25% was unsupported and retracted after the multi-seed data.



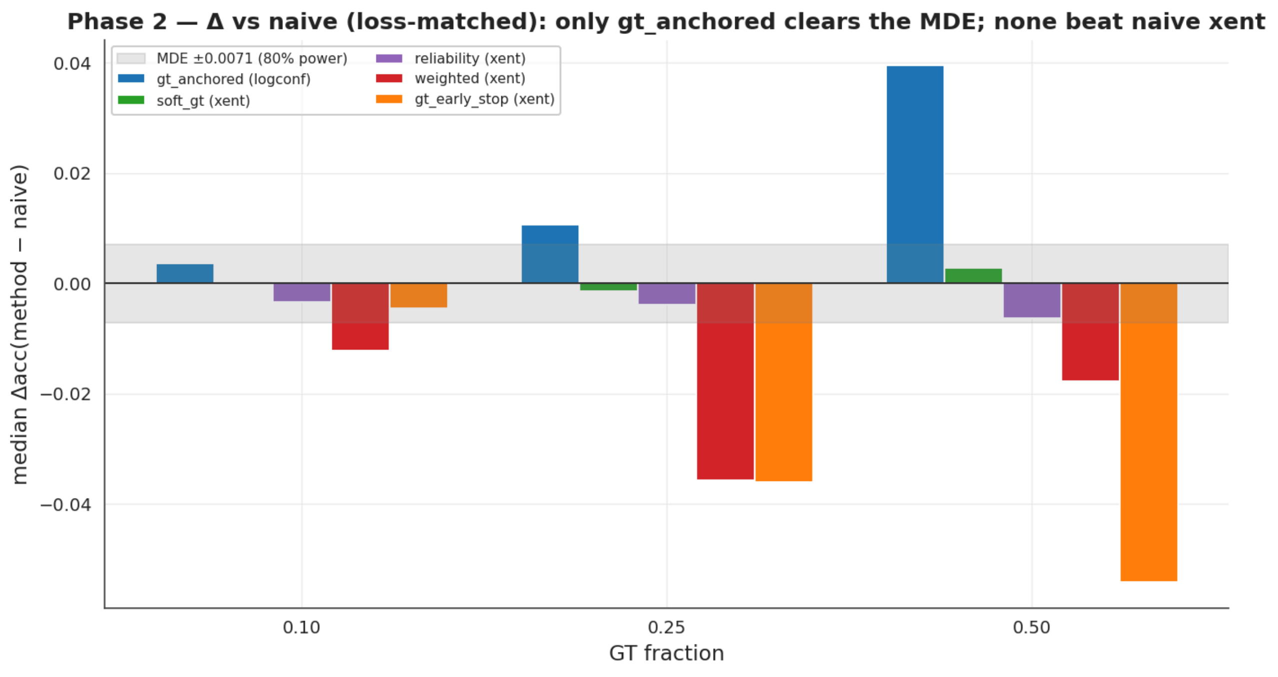
Where it's spent — within noise (allocation null)

- An oracle uses held-out GT to place the budget exactly on the teacher-wrong rows — an upper bound on any allocation rule.
- Paired against random placement at matched budget: oracle − random = +0.0006 (0.10), −0.0049 (0.25) — both below the MDE (0.0071) we could detect.
- If the best possible placement ties random, allocation methods have little to gain on this testbed.



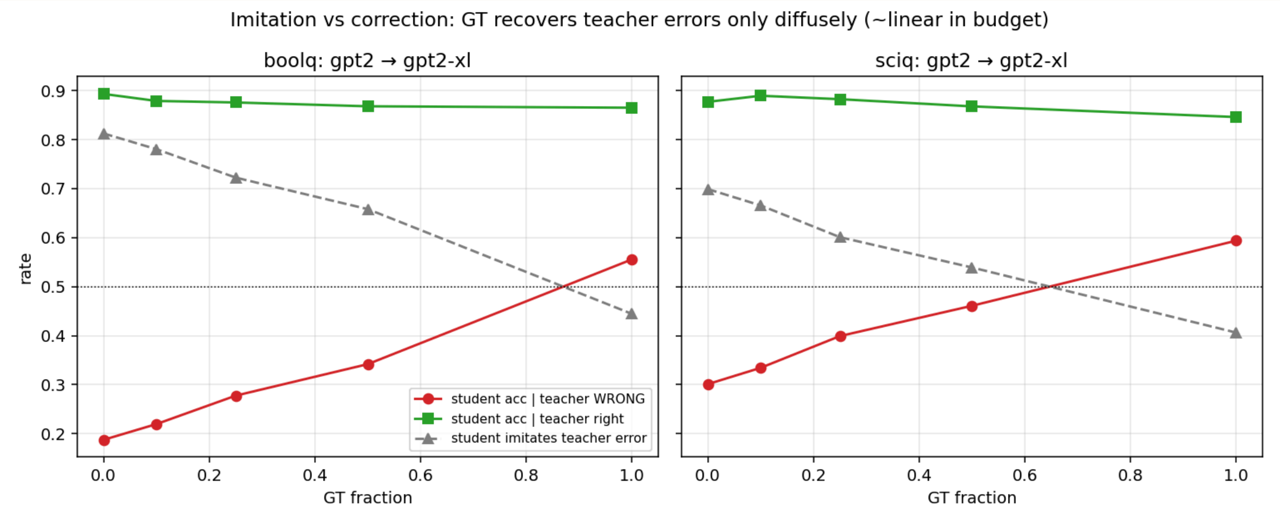
How it's combined — within noise (combination null)

- Five ways to use the GT rows more cleverly than plain mixing: (1) weight them more · (2) soften their labels · (3) exempt them from the confidence loss · (4) down-weight unreliable weak labels · (5) use them to choose when to stop training.
- Median Δ vs naive ≈ 0 across {0.10, 0.25, 0.50}; none shifts the curve left or raises the ceiling.
- Method 3 is the only one to clear the MDE (0.0071) — peaking at +0.040 at 0.50 — but without beating plain cross-entropy (0.642 vs 0.697).



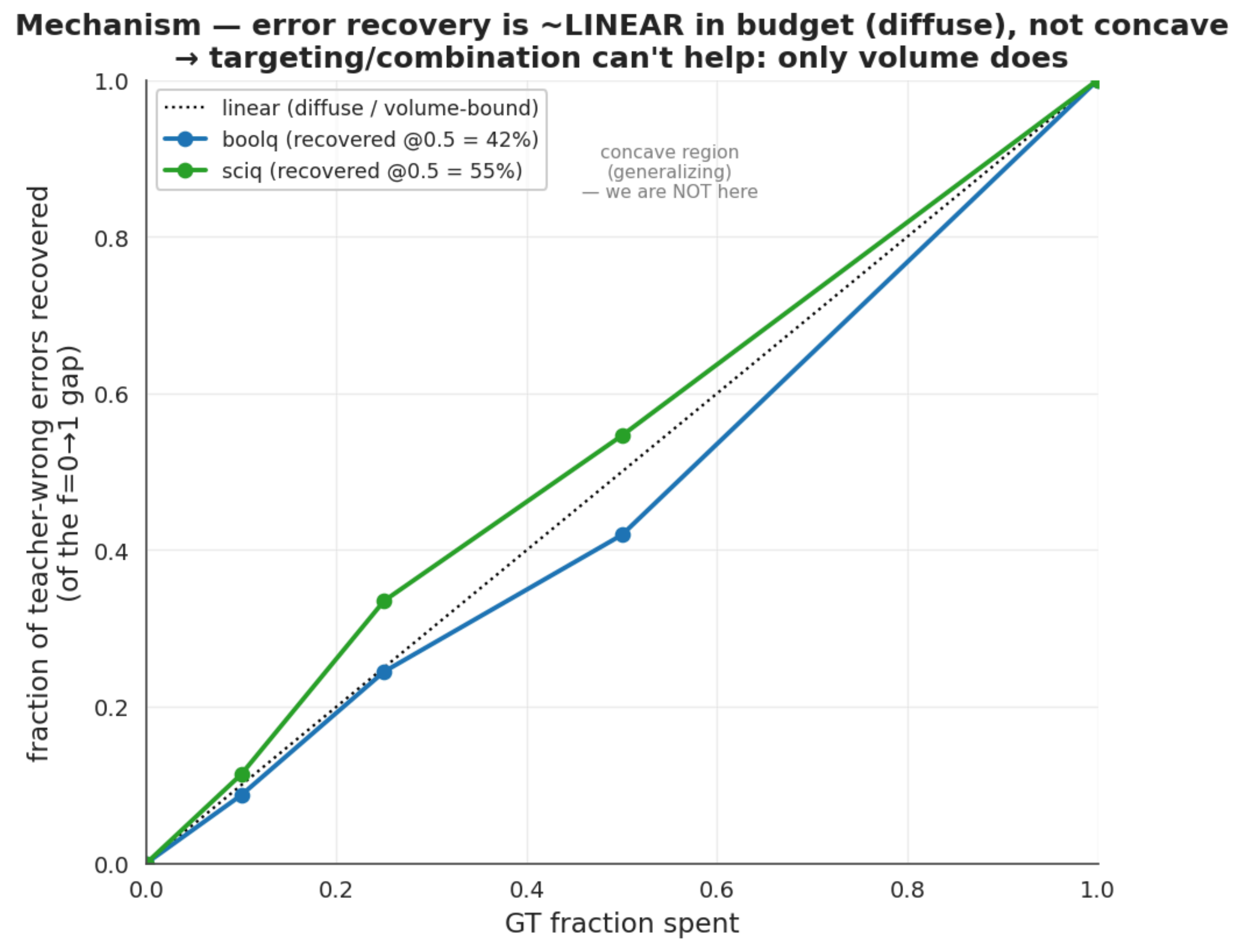
Mechanism: the student reproduces the teacher's errors

- Compared the gpt2 teacher and gpt2-xl student predictions, example by example.
- On teacher-wrong rows at 0% GT, the student reproduces the teacher's wrong answer 81% (BoolQ) / 70% (SciQ) of the time.
- Errors are largely inherited, not independent — the student adopts the teacher's specific wrong answers.



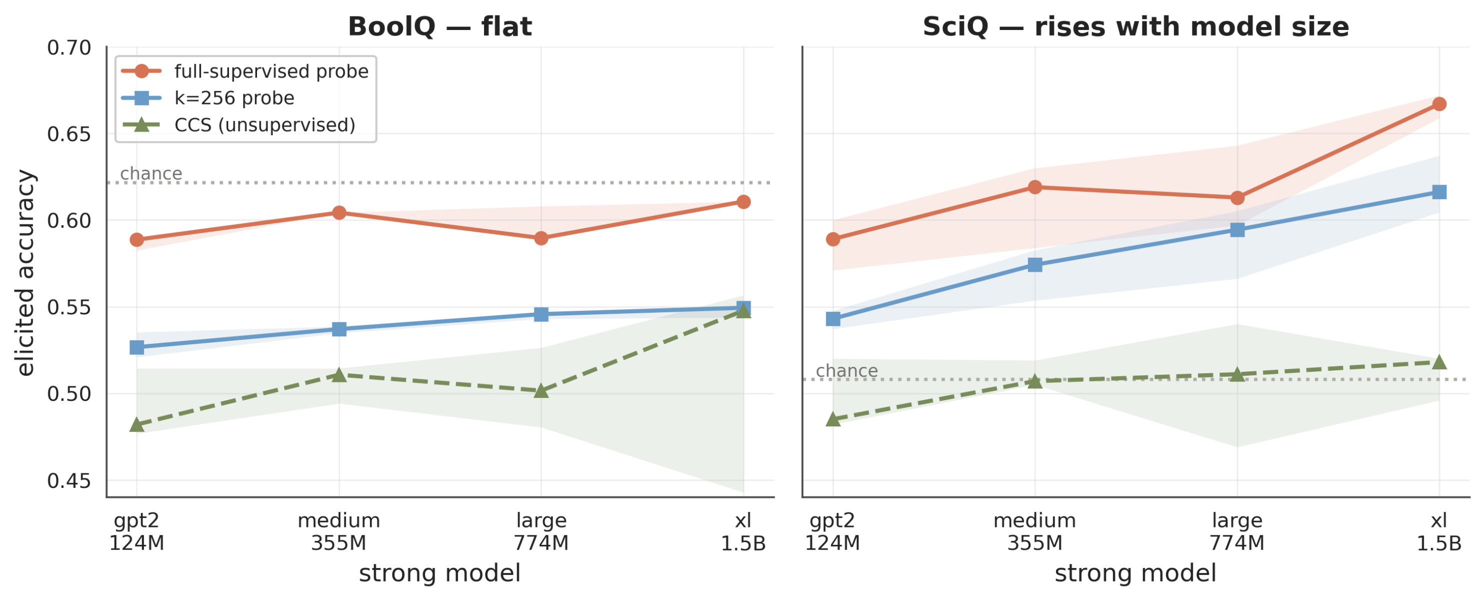
Mechanism: recovery is ~linear in budget (volume-dependent)

- Recovery of teacher-wrong rows is **~linear in budget** — 42% / 55% of the 0→100% gap recovered at 50%, tracking $y = x$ rather than a concave curve.
- A GT label's marginal value appears **roughly independent of which row it lands on**.
- This is the condition that produces both nulls: when recovery is volume-dependent, allocation (*where*) and re-weighting (*how*) have **little leverage**.



Extract the answer instead of supervising?

- The other lever is **elicitation** — read the answer out of the model's own internal state (no extra training), using the GT budget only to **point the readout the right way**.
- **Weak at GPT-2**: on BoolQ, even a classifier given all the labels barely beats guessing — but on SciQ it **rises with model size**.
- Elicitation may be impactful at a **larger capability gap** where the strong model genuinely knows the answer.



Prediction scorecard — hits and misses

Predictions for the fraction sweep — committed before the confirming seeds

	Prediction	Outcome
P1	knee at ~25%	✗ refuted — back-loaded, no knee (retracted)
P2	≤10% GT flat	✓ within noise
P3	mixing > GT-only	✓ (confound named, then controlled)
P4	confidence loss null	✓ inferior at every fraction
P5	scale interaction (gap → more GT)	— underpowered at GPT-2 scale
P6	0.25–0.50 plateau	✓ flat in that range

Predictions for the five combination methods — committed before the run: method 2 ✓ null, method 4 ✓ null (as flagged uncertain), **method 3** ◑ (strongest bet — rescues the confidence loss, still < cross-entropy), methods 1/5 ✗ negative.

Experiments across the three axes

Axis	Experiment	Outcome
How much	8-point fraction sweep, 0 → 1	back-loaded, saturates ~0.75
Where	error-targeting oracle + random control	null — within MDE
How	5 combination / loss methods	null — 1 above MDE, still < cross-entropy
Loss	cross-entropy vs confidence loss	confidence loss inert → dropped
Tasks	BoolQ + SciQ	replicates on both
Mechanism	teacher-vs-student prediction analysis	recovery ~linear in budget
Elicitation	read the answer out of the model (linear classifier + an unsupervised method)	weak at GPT-2; scales with model size (SciQ)
Robustness	8-seed variance; 5-seed baseline pass	exclusion by rule; 3-seed conclusions hold

Coverage spans the three axes, plus loss, task, robustness, and elicitation checks.

Takeaways

- Reproduced W2SG on the GPT-2 family and extended it to a supervision-budget setting.
- Decomposed the budget question into *how much / where / how* — a powered null on the last two, and a back-loaded, saturating curve on the first.
- One account — **inherited errors plus volume-dependent recovery** — is consistent with all three.
- So the binding constraint here looks like **scale**, not allocation or combination.
- A **promising initial result**: elicitation is weak at GPT-2 today, but the elicitable answer **grows with model size** — pointing to a larger-gap regime where the lever shifts to *eliciting* what the model already knows rather than *supervising* it.

Next steps

1. **Vary the capability gap.** Re-run the budget sweep with a bigger student–teacher gap (a weaker / handicapped teacher; larger families if allowed). If recovery turns from linear to **concave**, targeted GT starts to generalize — and *where* / *how* would matter again.
2. **Are the teacher's mistakes structured?** Train a **cheap** classifier — reusing the saved predictions — to find, from the model's own features, where the teacher is wrong. If it can't, the errors are scattered, with no pattern for allocation to exploit.
3. **Schedule the budget over training.** We mixed GT and weak labels in a fixed ratio throughout, never varying the *timing*. Compare GT-first vs gradually phased — recent work suggests the schedule can matter more than the loss.
4. **Iterate the loop.** Use the budget-trained student as the next teacher and repeat at the same total GT. Does a budget that's null in one pass add up over rounds?

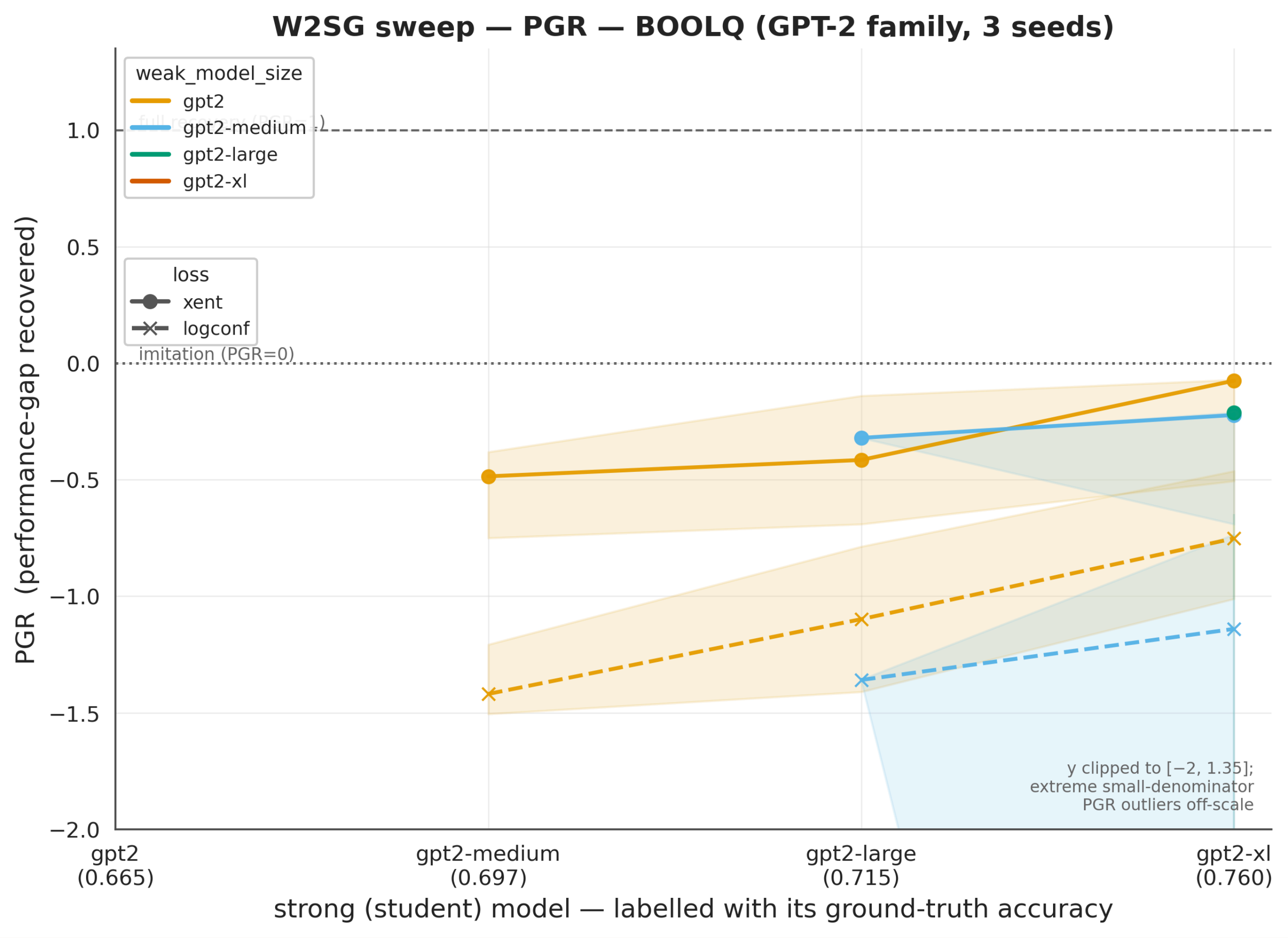
Ordered by relevance to the mechanism: (1) tests it directly · (2) explains it · (3–4) open new axes.

Appendix

frac=0 reproduction (acc + PGR) · transfer heatmap · per-method × pair detail · de-confound · cross-task

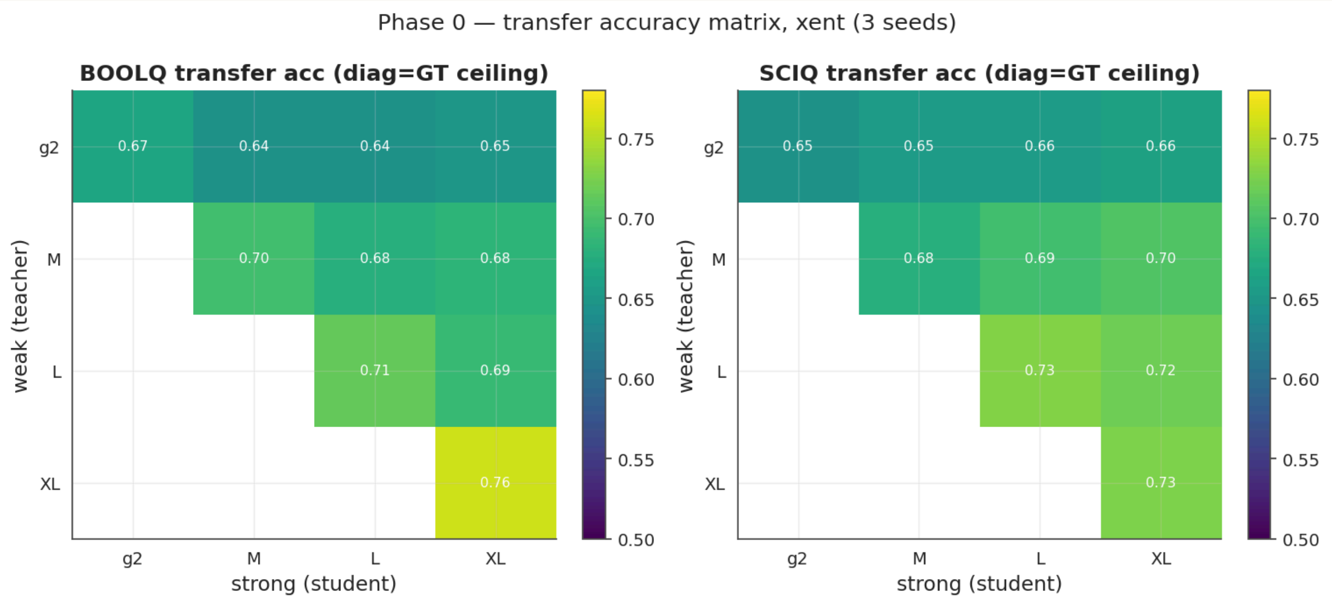
Appendix — frac=0 reproduction

- Standard W2SG sweep, GPT-2 family, BoolQ, **0% GT**, PGR axis.
- Median PGR **−0.27**; several pairs fall below the imitation line — BoolQ is low-signal at 0% GT.
- This is the baseline the 25%-GT sweep is measured against.



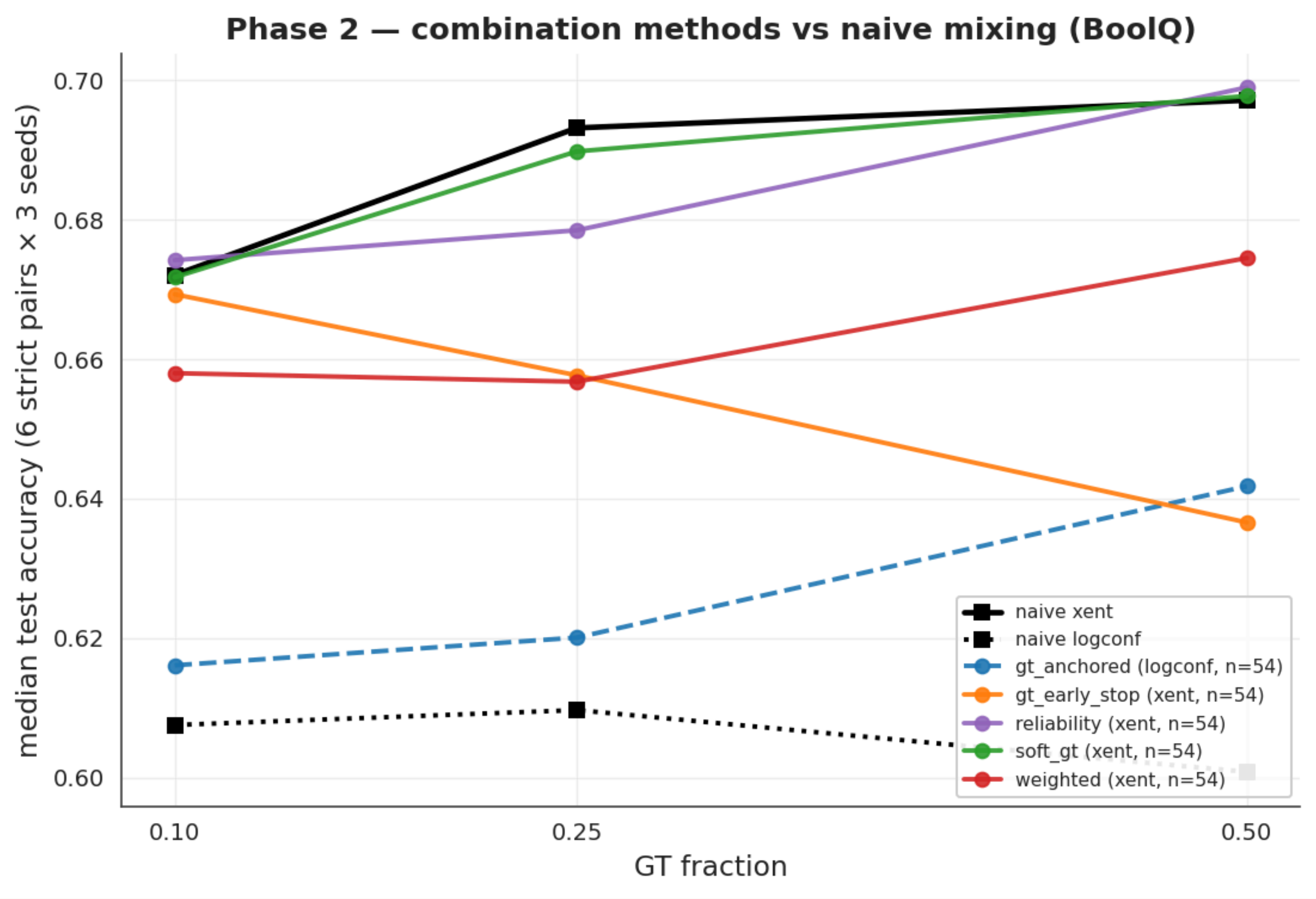
Appendix — transfer tracks the teacher more than student capacity

- With the teacher fixed, scaling the student adds ~+0.003; with the student fixed, a better teacher adds ~+0.04.
- Transfer tracks the teacher far more than student capacity — the static counterpart of the imitation result.



Appendix — combination portfolio, per-method detail

- All five methods track naive mixing across {0.10, 0.25, 0.50}.
- Method 3 (the confidence-loss variant) is the only one above the MDE; it rescues the confidence loss but stays under cross-entropy.



Appendix — weak labels carry information (de-confound)

- Ordering: **random labels < GT-only < naive mixing** (15/15 BoolQ, 18/18 SciQ pairs).
- Replacing weak labels with noise reduces accuracy → the mixing gain reflects weak-label information, not just added rows.